

TRIBAL INDIA -- L-62 CLASS NOTES

Paper 2 | Chapter 6.3 | Handout 3

Sarrthi IAS -- Anthropology Foundation (Nataraj Sir)

Displacement & R&R

Forest Policies

FRA 2006

Climate & TEK

Industrialization

Urbanization

Pencil = Handwritten class annotations included & marked with [Pencil]

Comprehensive Exam-Ready Notes -- UPSC CSE 2026

1. DEVELOPMENTAL PROJECTS & TRIBAL DISPLACEMENT

DISPLACEMENT OF TRIBALS

- **Context:** Infrastructure development necessary for nation, but tribals bear disproportionate cost
- **Causes:** Dams, Mining, Nuclear plants, Hydro-electric projects on forest lands
- **Scale:** 50M displaced; 40-50% are tribals (though only 8.6% of population)
- **Impacts:** Livelihood | Socio-cultural | Economic | Psychological
- **Measures:** NRRP 2007, FRA 2006, PESA 1996

Opening Context: Development of infrastructure is necessary for the growth of a nation. But not everyone benefits. Most projects on forest lands result in displacement of tribals living at the construction site, with a **huge impact** on their lives.

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Michael Cernea

Involuntary displacement should be **minimized**. If possible, **avoided**. If inevitable, it can be carried out such that the **livelihood of tribes is protected**. R&R should be "people centric" in approach.

1.1 Magnitude of Displacement

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Nalini Negi & Sutapa Ganguly

Study for University of Bielefeld, Germany (2011): ~**50 million people** displaced in India due to development projects -- of which **40-50% are tribals**.

Key Stat: Tribal population = **8.6%** of India, but projects displaced **over 50%** of tribals -- showing **disproportionate impact** of developmental policies on tribal communities.

1.2 Dam-Induced Displacement -- Case Studies

Polavaram Dam

Andhra Pradesh

Estimated displacement of **3 lakh people**; majority are tribals.

Sardar Sarovar Dam (Narmada)

Gujarat / MP / Maharashtra

12 Dec 1979: Despite widespread protests, Govt raised dam height. **30 major + 135 medium + 3000 small dams** constructed. Would provide water to **40M people** + irrigation + electricity. **2,00,000 displaced**, majority Adivasis.

1.3 Mining-Induced Displacement -- Case Studies

Pakri Barwadih Coal Mining Project

Hazaribagh District, Jharkhand | NTPC | North Karanpura Coalfield

- Land acquisition since **2016**: 3,319.42 ha (1,950.5 ha farmland + 643.9 ha forest + 101.22 ha settlements)
- 19 villages** face displacement (Barkagaon, Sonbarsa, Pakri Barwadih, Deoria Khurd)
- Compensation: Rs 9.85-10.85 lakh/adult -- but re-employed as **contract guards/labourers** on their own land
- Women's factory jobs ended abruptly when associated factories shut down; families left jobless
- Mining companies acquired **87,000+ ha** in Jharkhand (larger than Bengaluru)
- Most lands belonged to **STs and SCs**; skilled jobs go to **migrants from other states**, locals left unemployed
- Jharkhand's 2008 R&R Policy promised local benefit -- reality: **tribals lost both land & livelihood**

Chaal Coal Mine

Raigarh District, Chhattisgarh | SECL | Since 2003

- Displaced **~400 Kavar tribe families** from Lath village (only 56 left)
- Blasting & excavation destroyed farmlands & rice fields --> **forced migration**
- Mine waste into rivers --> **severe water contamination** --> lost clean water & fish (vital for subsistence)
- Rehabilitation promises remained **on paper only** -- villagers left without income or land
- Nearby: **Hindalco Industries** demolished Adivasi homes for mine expansion without consent/compensation
- Result: **land alienation + livelihood loss + erosion of tribal dignity & culture** -- once self-sufficient Adivasi villages became symbols of displacement

1.4 Impacts of Displacement (Detailed)

Note:

Displacement of tribes, especially **involuntary displacement**, has a huge impact not only at the socio-cultural level but also on the psychological state.

DIMENSION	KEY IMPACTS (DETAILED)
Livelihood	Tribes that had entire forest lands for themselves now have to move to the fringes with no access to forest or forest produce. They lose the natural habitat they have been living in for generations.
Socio-Cultural	a) Collectivism --> Individualism: Displacement reduces collective community values, shifting towards individualistic attitudes; loss of traditional community practices b) Agriculturalists --> Industrial Labour: Tribes forced to abandon traditional agricultural practices and become industrial labourers c) Religious Transformation: Slow fading away of many ritualistic cultures that were integral to tribal life d) Endogamy --> Exogamy: Marriage rules change -- many tribes shift from endogamy to exogamy due to displacement and mixing with non-tribal populations
Economic	a) Loss of Forest Produce: Significant source of livelihood cut off b) Landless Peasants: Traditional lands taken away for development projects; become landless c) Severe Poverty: Extreme poverty due to loss of traditional livelihoods & lack of access to resources d) Occupational Shift: Forced to abandon traditional economic occupations for industrial labour e) Inadequate Monetary Compensation: Remuneration for land acquisition is frequently insufficient and delayed , exacerbating economic hardships
Psychological	Insecurity, helplessness, identity crisis (covered in detail under industrialization section -- Das & Banerji framework)

1.5 Measures -- National R&R Policy, 2007

OBJECTIVE (6-POINT)

1. Minimize displacement; promote **non-displacing or least-displacing** alternatives

2. Ensure adequate R&R package with **expeditious implementation** & active participation of affected families

3. Special care for **SCs, STs, weaker sections** -- treated with concern & sensitivity

4. Provide **better standard of living**; concerted efforts for **sustainable income** to affected families

5. **Integrate rehabilitation concerns** into development planning & implementation process

6. Where displacement is on account of land acquisition: facilitate **harmonious relationship** between acquiring body & affected families through **mutual cooperation**

Conclusion: Displacement has always been a **companion of development**. But the process should be as **humane and voluntary** as possible. The focus of the government should be on **rehabilitation, not just displacement**. And the **cost of rehabilitation should be included in the project cost**.

2. FOREST POLICIES & TRIBALS

FOREST POLICY EVOLUTION

- **Colonial Era:** Control & exploitation (1865 --> 1878 --> 1894 --> 1927)
- **Post-Independence:** Gradual recognition (1952 --> 1980 --> 1982 --> 1988)
- **Rights Era:** Community governance (FRA 2006)
- **Shift:** Coercive control --> Participatory conservation

2.1 Tribal-Forest Relationship

The existence of tribal communities is **deeply linked** with forests. They share a **symbiotic relationship**, depending on forests for food, shelter, fuel, livelihood, and cultural identity.

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Tripathi (2016)

Tribes live in a "*harmonious and interdependent relationship*" with their forest environment. Their **economic life, social organisation, and magico-religious beliefs** all revolve around the forest ecosystem.

2.2 Colonial Period Forest Acts

ACT / POLICY	KEY PROVISIONS
Forest Act, 1865	Beginning of formal state control over forests; traditional practices restricted by law ; forests treated as state property rather than community resources
Forest Act, 1878	Expanded govt authority; divided forests into Reserved, Protected & Village Forests ; trespassing/grazing/collecting = offences ; consolidated Forest Department power over local use
First Forest Policy, 1894	Introduced " Eminent Domain " -- state has ultimate control over forest land; " <i>Rights and privileges of individuals must be limited</i> " in public interest; focused on revenue generation & timber supply for British needs
Indian Forest Act, 1927	Codified all previous forest laws into one statute; strengthened forest officers' power; Section 64 : authorised arrest without warrant for suspected violations; continued colonial legacy of exclusion & exploitation rather than conservation

2.3 Post-Independence Forest Policies

POLICY / COMMITTEE	KEY POINTS
National Forest Policy, 1952	Continued much of the colonial approach; limited recognition of tribal rights ; forests = national asset but ignored livelihood dependence; tribals given concessions, not rights -- maintaining their marginalisation
Natl Committee on Dev of Backward Areas (1980)	Recommended tribal rights & concessions be fully protected ; domestic needs met through forest produce; recognised importance of forest-based livelihoods for tribal well-being
B.K. Roy Burman Committee (1982)	Headed by eminent anthropologist B.K. Roy Burman ; tribes earn ~1/3 income from minor forest produce ; emphasised positive role of forest dwellers in preservation & regeneration; suggested national forest policy must integrate tribal participation for sustainable management
National Forest Policy, 1988	PARADIGM SHIFT: revenue-oriented --> people-oriented forest management Main objective: environmental stability & ecological balance Recognised symbiotic relationship of tribal people with forests Tribal communities to be actively involved in forest protection, regeneration & development

[Pencil] (Hiremath, 1994)

Introduced **Joint Forest Management (JFM)** -- partnership between Forest Dept & local communities

Recognised **MFP as a right** (not concession) of forest dwellers

Led to establishment of **TRIFED** (Tribal Cooperative Marketing Development Federation) to market forest produce & eliminate middlemen

3. FOREST RIGHTS ACT (FRA), 2006 -- Detailed Analysis

FRA 2006

- **Purpose:** Undo historical injustice to forest-dwelling communities
- **Two Functions:** (1) Legal recognition of rights (2) Community voice in conservation
- **3 Types of Rights:** Land | Use | Community (Protect & Conserve)
- **Central Authority:** Gram Sabhas
- **Special Provision:** PVTGs -- habitat & habitation rights [S. 3(1)(e)]
- **Linked Laws:** PESA 1996, FCA 1980, WLPA 1972

3.1 Objectives of FRA 2006

- Undo **historical injustice** occurred to forest-dwelling communities
- Ensure **land tenure, livelihood & food security** of forest-dwelling STs & OTFDs
- Strengthen conservation regime by including responsibilities & authority on forests -- **sustainable use, conservation of biodiversity & maintenance of ecological balance**

3.2 Conditions in Forest Areas (Section II)

Ground Reality in Forest Areas:

Because of the existing situation, **millions of people** are subject to **harassment, evictions** etc. in their own land. **Torture, bonded labour, extortion of money, sexual assault** are quite common among forest-dwelling communities.

3.3 Why Were People's Rights Not Respected? (Section III)

Analysis -- Root Cause:

The **Indian Forest Act, 1927** -- India's main forest law -- had little to do with conservation. It was created to serve the **British need for timber**. It sought to override customary rights and the forest management system by declaring **forests as state property** and exploiting their timber.

- A **single official** was to enquire into and "settle" disputes regarding land and forest rights
- These **all-powerful officials** unsurprisingly either did very little or recorded the rights of **powerful communities only**

Result of Colonial Forest Laws:

1. **Clearing of millions of hectares** of forest for **monoculture plantations** by the forest department
2. Forest laws **destroyed all community management & regulation systems** that had existed before, forcing people to choose between either **abandoning the forest entirely** or living as "**criminals**" within or near it

3.4 What Does the FRA Do? (Section IV)

TWO KEY FUNCTIONS OF FRA

1. Grants **legal recognition** to the rights of traditional forest-dwelling communities, **partially correcting** the injustice caused by forest laws
2. Makes a **beginning towards giving communities and the public a voice** in forest and wildlife conservation

3.5 Who Is Eligible? (Two-Stage Test)

STAGE CONDITIONS

Stage 1 (Both must be satisfied)	1. Primarily residing in forests or forest lands 2. Depend on forests and forest land for livelihood (namely "Bonafide livelihood needs")
Stage 2A	Above conditions true for 75 years --> "Other Traditional Forest Dweller" (OTFD)
Stage 2B	(i) Member of Scheduled Tribe + (ii) Residing in area considered as Scheduled Area --> "Forest Dwelling Scheduled Tribe"

3.6 Three Types of Rights (Key Provisions)

RIGHT TYPE	DETAILS
Land Rights	No rights to land not cultivated before 13 Dec 2005 & not currently cultivated. Without documents: claim up to 4 hectares (self-cultivation for livelihood only)
Use Rights	Collection of MFP (tendu, herbs, medicinal plants -- traditionally collected; no timber); grazing grounds & water bodies ; traditional areas of nomadic/pastoralist communities (communities that move with herds, as opposed to settled agriculture)
Right to Protect & Conserve (CFR)	First time this law gives community the right to protect and manage the forest Section 3(1)(i) : Community right to protect, regenerate, conserve & manage forest resources around villages CFRMCs established; but in several states, Forest Dept's control-oriented approach has slowed or obstructed CFR recognition, undermining community-based forest management
PVTG Rights	Section 3(1)(e) : Special habitat & habitation rights for PVTGs; Habitat-Area = common habitat + other habitats in reserved/protected forests

3.7 CFR -- Positive Case Studies

Chhattisgarh & Odisha: CFR Recognition

Kanger Ghati NP (CG) & Simlipal NP (Odisha)

CG recognised ~**4,000 CFRs** -- one of most proactive states.
Kanger Ghati = **2nd national park** (after Simlipal) with recognised CFR. In both parks, indigenous communities (e.g. Gudiapadar) allowed to reside inside park with **community-led regulations** for sustainable forest use. Model of **balance between conservation & livelihood rights** -- local participation strengthens forest governance.

Mendha Lekha & Betakata

Maharashtra & Odisha

Formed **CFRMCs** that sustainably regulate forest use, prevent illegal felling, and conserve biodiversity. **Model villages** for community stewardship of forests.

3.8 CFR -- Negative Case Study

Hasdeo Arand -- CFR Revocation (Conflict: Development vs Community Rights)

Ghatbarra village, Chhattisgarh | Adani Enterprises

- Ghatbarra: one of **first villages in CG** to receive CFR titles (**810 ha in 2013**)
- DLC **revoked CFR in 2016** for mining in PEKB coal block (Parsa East Kete Basen) -- operated by joint venture involving **Adani Enterprises**
- HABSS** (Hasdeo Aranya Bachao Sangharsh Samiti) challenged in court -- argued violation of **Gram Sabha consent** & FRA's core principle of recognising **pre-existing rights** (not granting new ones)
- CG High Court (Oct 2025): **upheld revocation** on technical grounds, disregarding evidence of Gram Sabha resolutions & community affidavits
- Activists: "**Mockery of FRA**" -- sets **dangerous precedent**: corporate interests override legally recognised community rights
- Exposes **gap between law and implementation** -- procedural loopholes & bureaucratic interpretations can reverse decades of progress in tribal & forest justice

3.9 Implementation Issues (14 Major Problems)

#	ISSUE	DETAILS & EXAMPLES
1	High Rejection Rates	Incomplete docs, no proof of pre-2005 cultivation, disputes between Forest & Revenue Depts; Odisha: 1.4L+ claims rejected ; most states not completed SC-ordered review (2019); genuine dwellers branded " encroachers "
2	Non-Communication	Rejected applicants rarely informed in writing -- violates Section 6(2) ; denies right to appeal within 60 days ; absence of clear timelines; irregular meetings of State Level Monitoring Committees
3	Weak District Support	District administrations fail to provide revenue/forest maps to Gram Sabhas; makes verification difficult --> gap between land claimed vs land granted
4	Faulty Evictions	Thousands evicted on incomplete or manipulated records ; e.g. Hasdeo Arand CFR title cancelled for coal mining, violating due process
5	Forest Dept Resistance	FD impedes CFR recognition & retains control; S. 3(1)(i) mandating community rights ignored in practice; Tamil Nadu: Kattunayakans (honey collectors) harassed by forest guards, prevented from collecting produce
6	Conflicting Laws	FRA conflicts with FCA 1980, WLPA 1972, CAMPA 2016 ; 2023 FCA amendment allows ecotourism & safaris on forest land without explicit Gram Sabha consent
7	Institutional Turf Wars	Poor coordination between MoTA & MoEFCC ; inter-ministerial friction delays implementation; contradictory directions at state level
8	Procedural Violations	Claims rejected by Forest Guards/Patwaris who have no legal authority to do so; Chittorgarh : oral rejections without notice, preventing any appeal
9	Data Manipulation	Chhattisgarh (ATREE report) : duplicate entries & inflated data; only 243 Gram Sabhas actually got CFR vs claimed 1,378
10	Lack of Awareness	Gram Sabha members & OTFDs lack awareness of FRA provisions; meetings postponed; claims delayed indefinitely; despite portals like Adi Prashikshan , awareness remains low in remote areas
11	Judicial Uncertainty	2019 SC eviction order : panic among 1.8M forest dwellers ; order stayed but most states yet to review rejected claims properly
12	Protected Area Displacement	1 lakh+ displaced from tiger reserves (NTCA relocations) without settling FRA claims ; reflects conservation-vs-rights conflict
13	Mining Violations	Forest clearances granted before FRA compliance ; Hasdeo Arand: CFR cancellation without Gram Sabha consent; activists call it " mockery of FRA "
14	Uneven State Progress	CFR recognition: Maharashtra 36%, Chhattisgarh 24%, Odisha 10% ; other major states nearly at zero -- showing administrative neglect

3.10 Significance & Achievements of FRA (11 Points)

#	ACHIEVEMENT	DETAILS
1	Historical Justice	Corrects colonial-era " historical injustice " faced by STs & forest dwellers; recognises both IFR & CFR/CFRR
2	Comprehensive Rights	Sections 3(1)(b), 3(1)(c), 3(1)(i) : legally empower communities to access, occupy, and manage forest lands, water bodies, grazing areas, and biodiversity resources
3	Empowered Gram Sabhas	Central authority for claim verification & forest governance --> participatory democracy at local level
4	Community Stewardship	Mendha Lekha (MH) & Betakata (OD) : CFRMCs sustainably regulate forest use, prevent illegal felling, conserve biodiversity
5	Economic Empowerment	Bamboo & NTFP rights raised incomes; Pajilibandha & Kalakani (Odisha) : women earn Rs.5,000-6,000/month via Van Dhan Kendras ; FRA links livelihoods with conservation

6	Ecological Benefits	Soura Kurlanda: restored degraded forests & increased biodiversity -- showing community control enhances conservation
7	Gender & PVTG Inclusion	Women & PVTGs participate in Van Suraksha Samitis , ensuring equitable access & decision-making power
8	Link with PESA	Odisha & MH merged FRA with PESA --> Gram Sabhas get legal corporate status under Article 243G , enabling self-rule
9	Tech Innovation	Gujarat (Narmada district): NGOs used GPS & satellite imagery for digital claim mapping; now replicated in other states
10	Admin Progress	MoTA 2024: ~60% claims recognised , covering 13 million acres ; steady progress in complex multi-tiered process
11	Global Model	When implemented properly: balances livelihood security with biodiversity conservation --> global model of justice-based environmental management

3.11 Way Forward & Suggestions (12-Point)

#	SUGGESTION	DETAILS
1	Review Rejected Claims	Transparent review; written reasons for rejection; ensure appeal rights under Section 6
2	Strengthen Gram Sabhas	Recognise as body corporates (MH & OD model) for fund management, decision-making & legal representation
3	Joint Coordination	Joint MoTA-MoEFCC mechanisms for consistent interpretation of forest & tribal policies
4	Digital Mapping	GPS & GIS mapping with BISAG & NRSC to eliminate duplication & data manipulation
5	Awareness & Training	Expand Tribal Research Institutes & Adi Prashikshan platforms for grassroots legal & procedural training of FRCs & community leaders
6	Livelihood Integration	Link CFR areas with TRIFED, Van Dhan Yojana, PM-JANMAN for forest-based income & self-reliance
7	Legal Safeguards	Mandate Gram Sabha consent before approving mining/diversion projects (prevent Hasdeo-type cases)
8	Inclusive Participation	Women, PVTGs, marginal groups in decision-making & forest management for inclusive governance
9	Monitoring & Transparency	State-level dashboards for claims data (filed, approved, pending); real-time accountability among district committees
10	Model Replication	Scale Mendha Lekha & Betakata ; promote peer learning via exposure visits across states
11	Policy Convergence	Integrate FRA with Green India Mission, CAMPA, National Biodiversity Strategy -- livelihood & conservation goals move together
12	National FRA Mission	Dedicated mission with separate budget & time-bound targets for claim settlement (proposed in Parliament discussions)

3.12 Supreme Court Order -- Feb 2019

SC Eviction Order (13 Feb 2019)

Affecting 16 states

- SC ordered **eviction of lakhs of STs & OTFDs** across 16 states whose FRA claims were rejected
- Bench directed chief secretaries to carry out evictions
- Eviction deadline: **24 July 2019**
- 28 Feb: Court **stayed its order** but warned "*the mighty & undeserving*" who encroached would get no mercy
- Stay due to role of **anthropologists & activists** raising awareness about **real reasons for high rejection rates + tribal protests**
- Court now examining whether **due process was followed** by Gram Sabhas & states before claim rejections

FRA Conclusion: The Forest Rights Act (2006) stands as a **landmark in ecological justice & tribal empowerment**, but its promise is yet to be fully realised due to **weak implementation, bureaucratic hurdles, and conflicting conservation priorities**. To achieve its full potential, India must **strengthen Gram Sabhas, ensure legal accountability**, and make **community-led forest governance** the foundation of sustainable forest management.

3.13 Transition: Colonial Forest Control --> Community Governance

#	TRANSITION DIMENSION	FROM --> TO
1	Control to Custodianship	State power & revenue extraction (1865-1927) --> tribal custodianship of forests (post-1988, 2006)
2	Management Model	Top-down bureaucracy --> Gram Sabhas as democratic forest managers (FRA 2006)

3	Philosophy	Extractive forestry --> rights-based environmental justice ; conservation without community participation is unsustainable
4	Purpose	Economic utility & timber --> ecological balance & JFM / CFR
5	Persisting Gap	Old administrative mindsets persist; MoTA vs MoEFCC conflicts; conservation vs livelihood priorities still unresolved
6	Way Ahead	Success depends on bridging historical contradictions ; tribal stewardship & ecological conservation must coexist through genuine decentralisation & capacity building

Forest Policy Conclusion: Forest policies in India reflect a journey from **colonial domination to democratic participation**. FRA (2006) embodies this transformation by recognising forest dwellers as **partners in conservation rather than intruders**. For its promise to be fulfilled: **effective implementation, community awareness, and strong Gram Sabha empowerment** so that justice, sustainability, and livelihood security truly reach those who live by the forest.

4. CLIMATE CHANGE & TRIBAL ECOLOGICAL KNOWLEDGE (TEK)

CLIMATE & TRIBALS

- **Paradox:** Strongest forest protectors <-- --> Severest climate impact victims
- **Sensitivity:** Livelihoods depend on forests, rainfall, soil fertility
- **Risks:** Erratic monsoons, shrinking forests, soil erosion, forest fires
- **Coping:** Traditional Ecological Knowledge (TEK)
- **Path Ahead:** Integrate rights (FRA+PESA) with climate adaptation

Context: India's tribal and indigenous communities are among the **strongest protectors of forests and biodiversity**, yet they face the **severest impacts of climate change**. Their livelihoods, food systems, and cultural traditions are being reshaped by ecological disruption. Tribal livelihoods are **highly climate-sensitive** because they depend on forests, rainfall, and soil fertility.

4.1 Risks & Problems Due to Climate Change

RISK	IMPACT ON TRIBALS
Erratic Monsoons & Droughts	Threaten cultivation & forest produce; Bhil (Gujarat) : once prayed for rain, now face destructive floods followed by prolonged dry spells
Shrinking Forests & Biodiversity Loss	Reduced access to MFP (tendu, mahua, bamboo) --> income & food security undermined
Soil Erosion & Deforestation	Worsens migration in Central India ; extreme heat & forest fires endanger both crops & health

4.2 Tribal Ecological Knowledge (TEK) -- Case Studies

TRIBE (LOCATION)	TEK PRACTICE
Jarawa (Andaman Islands)	Predict cyclones by observing fish movement & sea behaviour --> timely evacuation & disaster preparedness
Khasi (Meghalaya)	Community agroforestry & sacred groves --> preserve soil moisture & biodiversity in high-rainfall zones
Apatani (Arunachal Pradesh)	Rice-fish integrated farming in Ziro Valley --> conserve water & enhance soil fertility without chemicals
Dongria Kondh (Odisha) & Irular (Tamil Nadu)	Organic multi-crop farming with traditional pest control & seed-saving --> food & ecological security
Zabo System (Nagaland)	Combines rainwater harvesting + forestry + animal husbandry + agriculture --> maintains fertility & reduces drought stress

TEK Significance: These practices show how TEK **strengthens community resilience, conserves biodiversity, and supports sustainable adaptation** to climate change.

4.3 Continuing Struggles & Policy Disconnect

Despite their ecological wisdom, tribal communities remain marginalised in climate governance:

- CAMPA & Green India Mission: largely **top-down** --> led to "**green displacement**" through afforestation without community consent
- **Gender inequalities widening**: tribal women bear the burden of resource collection & loss of autonomy
- Traditional **rain forecasting & shifting cultivation cycles failing** as climate shifts become unpredictable --> eroding both livelihoods & ancestral knowledge
- **Institutional fragmentation** (MoTA vs MoEFCC); neglect of **Gram Sabha consent** under FRA weakens adaptation from below

4.4 Path Ahead: Integrating Rights & Resilience

Core Principle: A sustainable future demands that **traditional knowledge and modern science work together**.

- Integrate **FRA (2006) & PESA (1996)** into climate-adaptation frameworks; link with **PM JANMAN**
- Empower **Gram Sabhas** to manage eco-restoration & climate funds locally
- Document & scale up **Zabo & Apatani systems** across eco-regions
- Strengthen role of **tribal women & PVTGs** in decision-making & climate planning
- Blend **community forecasting** with modern **satellite-based early-warning systems** for better preparedness

Key Insight: The people who **nurture and protect** India's natural ecosystems are the ones **most endangered by climate change**. Their survival depends not only on external support but on recognising them as **equal partners in climate adaptation & forest governance**. Building climate resilience must begin with strengthening the **rights, knowledge, and institutions** of those who live by the forest.

5. INDUSTRIALIZATION & URBANIZATION -- IMPACT ON TRIBALS

INDUSTRIALIZATION & URBANIZATION

- **Context:** Tribal lands rich in forests AND mineral deposits
- **Promise vs Reality:** Development meant to uplift tribals -- instead uprooted them
- **Process:** Land acquisition --> meagre cash --> landless & moneyless
- **Problem:** Projects acquired MORE land than required --> private housing for outsiders
- **Impact (4D - Das & Banerji):** Social | Economic | Religious | Psychological
- **Positive:** Education, healthcare, art recognition, social reform

Opening Analysis -- The Broken Promise:

Tribal lands are not only rich in **forest resources** but also contain rich **mineral deposits**. The post-independence government, with development as the focus, tried to exploit the minerals for raw materials and constructed many projects in tribal lands. This development was **supposed to uplift the tribals, but that did not happen**. Instead, industrial and developmental projects have **uprooted the tribals from their dwellings**. They were deprived of their land and paid **meagre cash**, which they used up eventually. These projects made them **landless and moneyless**.

Land Acquisition Problem: Projects acquired **more land than required**, which is then used for **private housing** in many places, leaving original landowners destitute. Despite rules providing **90% reservation for locals** in employment & training, govt figures show **reality has been exactly the opposite**. Outsiders continue to get preferential treatment; land alienation continues uninterrupted. Tribals end up as **industrial/unskilled labour** living in outskirts slums.

[Pencil] HANDWRITTEN

Das and Banerji (1964)

Impact of industrialization & urbanization on tribals can be studied under **4 major dimensions -- social, economic, religious, and psychological** -- each reflecting deep structural changes in tribal society.

5.1 Case Studies -- Industrialization

Pakri Barwadih Coal Mining

Hazaribagh, Jharkhand | NTPC

3,319 ha acquired; 19 villages displaced; 87,000+ ha total in Jharkhand (> Bengaluru). Most lands belonged to **STs & SCs**. Skilled jobs to migrants from other states; locals left landless. Jharkhand's 2008 R&R Policy promised locals benefit first -- **reality: tribals lost both land & livelihood**.

Chaal Coal Mine

Raigarh, Chhattisgarh | SECL | Since 2003

400 Kavar families displaced from Lath (only 56 left); water contaminated by mine waste; promises on paper only; Hindalco demolished homes without consent. Once self-sufficient Adivasi villages became **symbols of displacement**.

5.2 Urbanization -- Why and How?

The Urbanization Process (Detailed Narrative):

Due to developmental projects, natives are **resettled elsewhere** -- losing access to forest, resources, agricultural land, **their whole livelihood**. To earn bread, they venture to the nearest town.

Poor tribals who formerly lived in a **secluded culture** must now adapt to a **cosmopolitan society** without access to education, equipment, or resources. There is an **identity issue**.

Life in town is also difficult -- people are forced to build and live in **filthy slums on the outskirts**. These slums lack basic amenities: **tap water, drainage, garbage disposal**.

Initially some got unskilled jobs, but once construction ended they were **unsuitable for skilled/semi-skilled jobs**. These jobs were taken by **outsiders who would eventually settle in tribal lands**, whereas tribals were driven to fringes as **"rejects"**.

5.3 Case Study -- Madhya Pradesh: Encroachment & Forest Loss

Madhya Pradesh -- Encroachment, Forest Loss & Dhyan Singh Case

Shahdol District, MP | 2022

- MP has India's **largest tribal population (~15%)** in two belts: Western (Bhil 49%) & Eastern (Gond 36%)
- Leads country in **forest land encroachment: 5,347 sq km lost** (highest in India)
- **Dhyan Singh case (2022)**: Tribal man committed suicide after forest dept demolished his hut for "encroaching" on **0.010 ha** -- barely a few metres
- **Irony**: He had already received money under **PMAY** to build that very house
- Dept justified action under **Section 36(1) of Indian Forest Act, 1927**
- For Dhyan Singh, it meant losing his **only shelter**

5.4 Four-Dimensional Impact Framework (Das & Banerji, 1964) [Pencil]

Context for the 4D Framework:

Industrialization & urbanization have been major agents of **socio-economic transformation** in post-independence India. While these processes created industries, infrastructure & employment, they also caused **widespread dislocation & cultural disintegration** among tribal communities. Tribal areas, being rich in minerals, forests & water, became focal points of development. However, these projects often **displaced the original inhabitants, disrupting their traditional way of life**.

Dimension 1: SOCIAL IMPACTS

SUB-AREA	DETAILED ANALYSIS
a) Loss of Home & Forests	Forests represent both livelihood and identity for tribals -- providing food, fuel, medicinal plants, shelter materials, and holding cultural & spiritual significance. Mining, dam & industrial projects have rapidly depleted these forests. Data: MP alone lost 219.7 ha of forest to infrastructure projects -- about 24% of India's total forest diversion over last decade (MoEFCC data). Disappearance of forests deprives tribals of economic base & sacred landscapes. Destruction of sacred groves & ecological systems weakens traditional beliefs & community bonds. Result: large-scale displacement, land alienation, loss of cultural stability.
b) Resources Exploited, Tribals Left Behind	Tribal regions contribute significantly to national economy via mineral extraction, but benefits seldom reach locals. Keonjhar district (Odisha) -- one of India's richest in iron ore & manganese: Despite mineral wealth, suffers severe malnutrition & poor health: - Over 13 lakh children stunted/underweight - 74.7% pregnant women anaemic - 25.9% women have low BMI (NFHS-5) Shows economic development primarily benefits corporations & outsiders; tribal inhabitants remain impoverished & marginalized.
c) Breakdown of Family & Kinship	Das & Banerji (1964): Industrialization & migration have weakened interpersonal ties within tribal families. Earlier: joint family system was norm -- collective land ownership & strong kinship relations. Migration --> fragmentation into nuclear units; weakening traditional authority structures; elders lose decision-making power; younger generation adopts individualistic values. Rise in divorces, inter-community marriages; decline in community festivals & rituals. Overall: shift from collective cultural life to isolated urban existence; erosion of kinship solidarity & community cohesion.

Dimension 2: ECONOMIC IMPACTS

SUB-AREA	DETAILED ANALYSIS
a) Shift in Occupation	Before industrialization: subsistence agriculture, hunting, fishing, forest collection; family = basic unit of production. Industrialization introduced individual wage labour in factories, mines, construction sites. Changed not only mode of production but also nature of work: Factory employment = rigid hours, repetitive tasks, impersonal relations -- contrasting with flexible & communal nature of traditional tribal work. Tribals, accustomed to seasonal agricultural cycles, find industrial work monotonous & emotionally draining. Das & Banerji: Labour shifted from being a "collective social activity" to an "individual economic necessity".
b) Unemployment & Casual Labour	Despite promises of local employment, industrial projects rarely provided sustainable jobs to displaced families. Majority of skilled & semi-skilled positions occupied by non-tribal workers; tribals employed temporarily as unskilled labourers. After project completion: jobs vanish -- families left without income or land. Migration to urban centres increases: tribals end up as construction workers, street vendors, domestic

helpers -- living in poor settlements **without sanitation, clean water, or healthcare**. Represents movement from **self-sufficient agrarian lifestyle** to **economic dependency & urban poverty**.

Dimension 3: RELIGIOUS IMPACTS

Industrialization & urbanization led to **gradual decline in tribal religious & ritual life**. Rituals connected to **agriculture, harvest, and forest worship** are now either shortened or discontinued.

Das & Banerji (1964): Simplification of traditional ceremonies occurred largely due to **financial hardship** and **lack of leisure** in urban/industrial settings.

Community shrines & sacred groves often inaccessible after displacement -- weakens traditional belief systems. Absence of **ritual spaces & collective participation** further contributed to breakdown of **communal and spiritual identity** among displaced tribals.

Dimension 4: PSYCHOLOGICAL IMPACTS

Psychological consequences are both **severe and long-lasting**. Displacement, loss of land, cultural alienation create deep sense of **insecurity & helplessness**.

Individuals who once lived as **independent cultivators/forest gatherers** are now **dependent on employers/contractors**. Resulting **loss of control over their own lives** leads to **low self-esteem, frustration, and anxiety**.

Das & Banerji: Described this as emergence of a sense of "**powerlessness**" among tribal populations.

Younger generation, **disconnected from traditional culture**, faces **identity crises** and higher vulnerability to **substance abuse**. These conditions signify breakdown not only of economic security but also of **psychological well-being**.

5.5 Positive Impacts

AREA	POSITIVE OUTCOME (DETAILED)
Education, Healthcare & Infrastructure	Industrialization indirectly facilitated the spread of education, healthcare, and infrastructure in several tribal regions
Social Reform	Exposure to new ideas led to gradual social reform : declining acceptance of witch-hunting, child marriage, and gender-based taboos
Art & Craft Recognition	Traditional art forms gained national & international recognition : Gond paintings (MP) & Warli art (MH) became symbols of indigenous creativity; promoted through cultural organisations & cooperatives
Economic Empowerment	Several NGOs & self-help groups assist tribal communities in marketing handicrafts, forest produce & traditional goods -- enabling economic empowerment & cultural preservation

Conclusion: Industrialization & urbanization have brought both **development and destruction** to India's tribal societies. While these processes expanded infrastructure and created new economic opportunities, they also **displaced millions, dismantled traditional social systems, and eroded cultural values**.

The need of the hour: ensure **inclusive and participatory development** where tribal communities are treated as **partners rather than victims**. Sustainable development must integrate **economic growth with cultural preservation** and guarantee that the rights and welfare of tribal populations are **not compromised in the pursuit of modernization**.

APPENDIX: ALL HANDWRITTEN ([Pencil]) ANNOTATIONS

#	SCHOLAR / NAME	CONTEXT & CONTRIBUTION	PAGE
1	Michael Cernea [Pencil]	Involuntary displacement should be minimized/avoided; R&R must be "people centric"	Pg 3
2	Nalini Negi & Sutapa Ganguly [Pencil]	Study for Univ. of Bielefeld (2011): 50M displaced, 40-50% tribals	Pg 3
3	Tripathi (2016) [Pencil]	Tribes live in "harmonious & interdependent relationship" with forests; economic life, social organisation, magico-religious beliefs revolve around forest ecosystem	Pg 7
4	Hiremath (1994) [Pencil]	Tribals should be actively involved in forest protection, regeneration & development (NFP 1988 context)	Pg 8
5	Das & Banerji (1964) [Pencil]	4 dimensions of industrialization/urbanization impact: social, economic, religious, psychological; "emergence of a sense of powerlessness"; labour shifts from "collective social activity to individual economic necessity"	Pg 23

LAST-MINUTE CHEAT SHEET -- L-62 (Chapter 6.3)

Scan this in 5 minutes before the exam -- every topic, scholar, case study & key stat on one spread

TOPIC	MAIN POINTS (KEYWORDS)	KEY SCHOLARS	KEY CASE STUDIES
Displacement & R&R	* Tribals = 8.6% pop but 50%+ displaced * 50M displaced; 40-50% tribals * Dams, mining, nuclear, hydro-electric * Impacts: livelihood loss, collectivism-->individualism, endogamy-->exogamy, landless peasants, ritual fading, inadequate compensation * NRRP 2007: minimize displacement, people-centric, active participation, sustainable income, integrate rehabilitation into project cost	[Pencil] Michael Cernea -- minimize/avoid displacement; "people centric" R&R [Pencil] Nalini Negi & Sutapa Ganguly -- Bielefeld study (2011): 50M displaced, 40-50% tribals	Polavaram Dam (AP) -- 3L displaced Sardar Sarovar/Narmada -- 2L displaced, majority Adivasis, 30+135+3000 dams Pakri Barwadih (Jharkhand) -- NTPC, 19 villages, 3319 ha, 87K+ ha total Chaal Mine (CG) -- 400 Kavar families, water contamination, Hindalco
Forest Policies	* Symbiotic tribal-forest relationship * Colonial: 1865 (state control) --> 1878 (Reserved/Protected/Village) --> 1894 (Eminent Domain, "rights must be limited") --> 1927 (codified, S.64 arrest w/o warrant) * Post-1947: 1952 (concessions not rights) --> 1980 (protect tribal rights) --> 1982 (BK Roy Burman: 1/3 income from MFP) --> 1988 (PARADIGM SHIFT: JFM, MFP=right, TRIFED, people-oriented) * Transition: coercive control --> participatory conservation	[Pencil] Tripathi (2016) -- "harmonious & interdependent" tribal-forest relationship [Pencil] Hiremath (1994) -- tribals actively involved in forest protection (NFP 1988) B.K. Roy Burman (1982) -- 1/3 income from MFP; positive role of forest dwellers	Colonial: forests = state property for British timber; destroyed community systems; people = "criminals" NFP 1988 --> JFM model TRIFED -- market forest produce, eliminate middlemen
FRA 2006 -- Rights	* Undo historical injustice * 2 functions: legal recognition + community voice in conservation * Forest conditions: harassment, evictions, torture, bonded labour, sexual assault * Why rights not respected: single official "settled" disputes; monoculture plantations; community systems destroyed * 3 rights: Land (pre-13 Dec 2005, max 4 ha) Use (MFP, grazing, no timber) Community (protect & manage, S.3(1)(i)) * PVTG habitat under S. 3(1)(e) * Eligibility: 75 yrs (OTFD) OR ST+Scheduled Area * Gram Sabha = central authority	--	+ Mendha Lekha (MH) -- model CFRMC + Betakata (OD) -- community stewardship + Kanger Ghati NP (CG) -- 2nd NP with CFR + Simlipal NP (OD) -- 1st NP with CFR + Gudiyapadar -- community-led regulations inside NP
FRA 2006 -- Issues	* 14 implementation issues * High rejection rates (Odisha: 1.4L+ rejected); branded "encroachers" * Non-communication of rejections (violates S.6(2)) * Forest Dept resistance; MoTA vs MoEFCC turf war * Conflicting laws: FCA 1980, WLPA 1972, CAMPA 2016 * Data manipulation (ATREE: 243 vs claimed 1378) * SC 2019 eviction order (stayed) -- 1.8M at risk * 1L+ displaced from tiger reserves (NTCA) * CFR progress: MH 36%, CG 24%, OD 10%, rest ~ 0% * MoTA 2024: ~60% claims done, 13M acres	--	X Hasdeo Arand (CG) -- Ghatbarra 810ha CFR revoked for Adani mining (PEKB); HABSS challenged; HC upheld; "mockery of FRA" X Kattunayakans (TN) -- honey collectors harassed X Chittorgarh -- oral rejections, no appeal
FRA -- Way Forward	* Review rejected claims; written reasons; appeal under S. 6 * Gram Sabhas as body corporates (MH/OD model) * Joint MoTA-MoEFCC coordination * GPS/GIS mapping (BISAG, NRSC) * Link with TRIFED, Van Dhan, PM-JANMAN * Mandate Gram Sabha consent before mining * National FRA Mission with separate budget * FRA = landmark in ecological justice; community-led forest governance as foundation	--	+ Pajilibandha & Kalakani (OD) -- women earn Rs.5-6K/month via Van Dhan Kendras + Soura Kurlanda -- restored degraded forests + Gujarat (Narmada) -- GPS/satellite claim mapping

Climate Change & TEK	* Tribals = strongest protectors yet worst victims * Highly climate-sensitive (depend on forests, rainfall, soil) * Erratic monsoons, biodiversity loss, soil erosion, forest fires * TEK = Traditional Ecological Knowledge * TEK strengthens resilience, conserves biodiversity, supports sustainable adaptation * Policy disconnect: CAMPA/GIM = top-down --> "green displacement" * Gender inequalities widening; rain forecasting failing * Integrate FRA+PESA into climate adaptation; empower Gram Sabhas; scale TEK models; tribal women + PVTGs in climate planning	-- Jarawa (Andaman) -- predict cyclones via fish/sea Khasi (Meghalaya) -- sacred groves & agroforestry Apatani (Arunachal) -- rice-fish farming, Ziro Valley Dongria Kondh (OD) / Irular (TN) -- organic multi-crop Zabo (Nagaland) -- rainwater + forestry + husbandry Bhil (Gujarat) -- floods then drought
Industrialization & Urbanization	* Mineral-rich tribal lands --> mining --> displacement * Projects acquire MORE land than required --> private housing * 90% local reservation rule exists but outsiders get jobs * Tribals = landless & moneyless; "rejects" in outskirts slums * 4 impacts (Das & Banerji 1964): 1. Social: home/forest loss (MP: 219.7ha = 24% India's diversion), kinship breakdown (joint-->nuclear), Keonjhar mineral wealth vs malnutrition 2. Economic: subsistence-->wage labour (collective-->individual); unemployment; construction workers/vendors/helpers 3. Religious: rituals shortened/dropped; shrines inaccessible; financial hardship & lack of leisure 4. Psychological: powerlessness, identity crisis, substance abuse, loss of control * Positive: education, health, social reform, Gond/Warli art, NGOs/SHGs, cooperatives	[Pencil] Das & Banerji (1964) -- 4-dimension framework; "collective social activity --> individual economic necessity"; "emergence of powerlessness" Pakri Barwadih (Jharkhand) -- 87K+ ha; skilled jobs to migrants Chaal Mine (CG) -- Kavar tribe; SECL Keonjhar (OD) -- iron ore rich but 13L children stunted, 74.7% pregnant women anaemic, 25.9% low BMI Dhyan Singh (MP, 2022) -- suicide; hut demolished for 0.010 ha; had PMAY funds; S.36(1) IFA 1927 Gond paintings (MP), Warli art (MH) -- positive

SCHOLAR QUICK-RECALL (All [Pencil] = Handwritten)

SCHOLAR	YEAR	ONE-LINER TO WRITE IN EXAM
Michael Cernea [Pencil]	--	"Involuntary displacement should be minimized; R&R must be people-centric"
Nalini Negi & Sutapa Ganguly [Pencil]	2011	"50M displaced in India; 40-50% are tribals" (Bielefeld University study)
Tripathi [Pencil]	2016	"Tribes share a harmonious and interdependent relationship with forests" -- economic, social, magico-religious beliefs revolve around forest
Hiremath [Pencil]	1994	"Tribal communities should be actively involved in forest protection, regeneration & development" (NFP 1988 context)
Das & Banerji [Pencil]	1964	"4D impact: social, economic, religious, psychological"; "collective social activity --> individual economic necessity"; "emergence of powerlessness"
B.K. Roy Burman	1982	"Tribes earn ~1/3 income from MFP; forest dwellers play positive role in preservation"; integrate tribal participation in forest management

KEY NUMBERS TO REMEMBER

NUMBER	WHAT IT REFERS TO
8.6%	Tribal share of India's population -- but displaced disproportionately (50%+)
50 million	Total people displaced by development projects in India (Negi & Ganguly, 2011)
40-50%	Share of displaced who are tribals
3 lakh	Displaced by Polavaram Dam (AP)
2 lakh	Displaced by Sardar Sarovar Dam (Narmada); 40M people get water
87,000+ ha	Land acquired by mining cos in Jharkhand (larger than Bengaluru)
3,319 ha	Pakri Barwadih project land (1950.5 ha farm + 643.9 ha forest + 101.22 ha settlements)
400 families	Kawar tribe displaced from Lath village by Chaal Mine (only 56 left)
1/3 income	Tribal income from MFP (B.K. Roy Burman, 1982)
4 hectares	Max land claimable under FRA without documents
13 Dec 2005	FRA cut-off date -- must have cultivated before this
75 years	Residence proof needed for OTFD status under FRA
810 ha (2013)	Ghatbarra CFR title -- revoked in 2016 for mining
4,000 CFRs	Recognised by Chhattisgarh
60% / 13M acres	FRA claims recognised so far (MoTA 2024)
1.8 million	Forest dwellers at risk from SC 2019 eviction order
1 lakh+	Displaced from tiger reserves (NTCA) without settling FRA claims
219.7 ha	MP forest lost to infra projects (24% of India's total diversion)
5,347 sq km	Forest land encroachment in MP -- highest in India
13L children	Stunted/underweight in Keonjhar (OD) despite mineral wealth
74.7%	Pregnant women anaemic in Keonjhar (NFHS-5)
0.010 ha	Land for which Dhyani Singh's hut was demolished (MP, 2022)

KEY LAWS, POLICIES & SCHEMES -- Quick Reference

LAW / POLICY / SCHEME	YEAR	REMEMBER AS
Forest Act	1865	First state control over forests; traditional practices restricted
Forest Act	1878	Reserved / Protected / Village forests; trespassing/grazing = offences
First Forest Policy	1894	"Eminent Domain" -- state ultimate control; "rights must be limited"; revenue & timber for British
Indian Forest Act	1927	Codified all laws; S. 64 arrest w/o warrant; S. 36(1) used to demolish tribal huts; had little to do with conservation -- served British timber needs
National Forest Policy	1952	Concessions, not rights -- continued colonial mindset
Natl Cttee on Dev of Backward Areas	1980	Protect tribal rights; domestic needs from forest produce
B.K. Roy Burman Committee	1982	1/3 income from MFP; positive role of forest dwellers; integrate tribal participation
National Forest Policy	1988	PARADIGM SHIFT -- JFM, MFP as right, TRIFED, people-oriented, symbiotic relationship
PESA	1996	Self-governance for Scheduled Areas; linked with FRA
FRA	2006	3 rights (land/use/CFR); undo historical injustice; Gram Sabha central; PVTGs S.3(1)(e)
NRRP	2007	Minimize displacement; people-centric R&R; rehabilitation cost in project cost
Jharkhand R&R Policy	2008	Promised locals benefit first -- failed in reality
CAMPA	2016	Compensatory afforestation -- conflicts with FRA
FCA Amendment	2023	Allows ecotourism/safaris without Gram Sabha consent
PM-JANMAN / Van Dhan / TRIFED	--	Livelihood schemes linked with FRA/CFR areas
PMAY	--	Housing scheme -- Dhyan Singh had PMAY money but hut demolished

Source: Sarrthi IAS -- Anthropology Foundation -- Tribal India Handout 3 (L-62) | Nataraj Sir
Paper 2 | Chapter 6.3 | UPSC CSE 2026

[Pencil] = Handwritten class annotations (filled blanks from class)